

Advanced Communication Architectures (TSM_AdvComArc)

SDN using EVE (Emulated Virtual Environment)

Topics: EVE (Emulated Virtual Environment)
Estimated time: ca. 2.0 Hours

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Installing and using the SDN emulation environment EVE

Please consult the document “EVE-NG Community Cookbook”, explaining EVE-NG (Emulated Virtual Environment – Next Generation).

Download the installer file for the EVE Community edition and install it on a virtual environment.

If using MS Windows, also download the appropriate toolkit from the website.

Please install the environment in a virtual machine.

While installing EVE, please look for the differences between EVE and another alternate solution: GNS3, and document the findings:

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The VM should be reachable from the hosting computer, therefore please pay attention to its IP configuration and to the network configuration of the virtualization environment.

Once installed, please launch the EVE control panel from a browser.

Take a look at the tools and configurations made available by the program.

Please check the infrastructure libraries made available by EVE.

Please check the website of ARISTA, a provider of SDN solutions. Try looking for libraries and extensions that will allow enriching the palette offered natively by EVE.

Try implementing the following architecture, focusing only on L3 devices and omitting elements belonging to lower OSI layers (basically the 4 routers):

Once the system will be successfully implemented, run some simulation campaigns.

Please try interconnecting the 2 EVE implementations, define border gateways in order to allow the full system to work, and test the system.

Please indicate some physical routers, at least 1 Juniper and 1 CISCO router that may be connected to the system.

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